

-2 cm air entry option in the van Genuchten-Mualem soil hydraulic functions

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Questions often arise about the meaning and importance of the -2 cm option when selecting the van-Genuchten-Mualem (VGM) hydraulic function in the HYDRUS codes. The option is identified in the “Soil Hydraulic Property Model” window under “van Genuchten-Mualem” and then the subheading “With Air-Entry Value of -2 cm”. This option is recommended only for relatively fine-textured soils when the VGM parameter n becomes very close to its lower limit of 1.0 (e.g., less than 1.1 or 1.2). Selecting this option will introduce a small correction (like an air-entry value) in the water retention function to force the slope of the retention function ($d\theta/dh$) to become zero when approaching saturation at $h=0$.

The correction has absolutely no effect on the macroscopic description of the water retention function of fine-textured soils. But it substantially changes the shape of the hydraulic conduction function, $K(h)$, as predicted with the VGM statistical pore-size distribution model. The predicted $K(h)$ is then far less nonlinear and much more accurate for fine-textured soils (typically clays and silty clays). It may also avoid numerical problems when modeling ponded infiltration or other variably-saturated flow problems. The -2 cm correction was first suggested by Vogel and Cislerová (1988) and worked out further by Vogel et al. (2001). Schaap et al. (2006) later showed that a -4 cm correction would be slightly better than -2 cm, but the difference is very minimal.

The -2 cm correction should not be used for medium-, and especially coarse-textured, soils when the value of VGM parameter α becomes relatively large (say >0.02 or 0.05 cm). This rule applies unless n is again less than about 1.1 or 1.2, which generally is not the case for medium- and coarse-textured soils. The -2 cm should then be avoided since it introduces a Brooks and Corey (1964) type air-entry correction into the retention function to make the function less accurate for many medium- and coarse-textured soils (Schaap and van Genuchten, 2006).

References

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